

Effects of Intervention on Compliance to Referral and Lifestyle Recommendations Given at Cholesterol Screening Programs

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To enhance compliance to physician referral as well as dietary and other lifestyle recommendations given at blood cholesterol (BC) screening programs, we randomized Pawtucket Heart Health Program SCORE (screening, counseling, referral event) participants with elevated BC levels into one of four groups: usual care group; a participant intervention group (mailed reminder letter and refrigerator magnet); a physician intervention group (mailed packet to participant's physician including letter, National Cholesterol Education Program [NCEP] guidelines, and preaddressed postcard to mail to patient); and a group that received both interventions. Beginning four months after the screening, we surveyed study subjects by phone. The participant intervention increased recall of physician referral and dietary recommendations; however, neither intervention successfully improved compliance to referral or dietary and lifestyle recommendations. Overall, 58%, 67%, and 34% of subjects reported

complying to physician referral, dietary recommendations, and lifestyle recommendations, respectively. Referral compliance was associated with a longer time interval between screening and survey (relative risk [RR] = 1.3, 95% confidence interval [CI] = 1.0, 1.7), possession of medical insurance that covered physician visits (RR = 2.1, 95% CI = 0.98, 4.4), and history of hypertension (RR = 2.6, 95% CI = 1.1, 5.8). Dietary compliance was positively associated with baseline BC levels ≥ 240 mg/dL (RR = 3.3, 95% CI = 1.4, 7.3) and negatively associated with increasing age; each one year increase in age corresponded to a 3% decrease in compliance (RR = 0.97, 95% CI = 0.9, 1.0). More research is needed to determine the barriers to referral compliance, the best ways to encourage subjects to comply with referral recommendations, and the most cost-effective methods to improve compliance to dietary and other lifestyle recommendations. [Am J Prev Med 1994;10:275-82]

An elevated blood cholesterol (BC) level is an important risk factor for coronary heart disease (CHD).¹⁻³ Because approximately 40% of American adults have BC levels that require medical intervention,⁴ reducing the prevalence of this risk factor is a public health priority.^{3,5,6} The purpose of public BC screenings is to identify and educate individuals with high BC levels and refer them to physicians for confirmatory testing and treatment.^{7,8} Unfortunately, many individuals who attend public BC screening programs and are referred to their physician for further testing and treatment for a high BC level do not follow this referral advice.⁹⁻²³ Several investigators have introduced methods to enhance compliance to referral advice given at BC screen-

ings;²⁴⁻²⁷ however, only one was successful,²⁵ and none studied referral compliance of individuals with BC levels that met the 1988 National Cholesterol Education Program (NCEP) guidelines for physician referral.^{3,8} Because referral compliance following BC screenings has been poorly investigated, this issue needs further research.^{28,29}

The Pawtucket Heart Health Program (PHHP), a National Heart, Lung, and Blood Institute (NHLBI) heart disease prevention research project,^{30,31} has been conducting community-based BC Screening, Counseling, Referral Events, (SCOREs) for eight years.^{11,32,33} These BC SCOREs incorporate the national guidelines for public BC screening⁸ and have served as a model for other BC screening programs. The importance of getting individuals with elevated BC levels into medical treatment prompted PHHP to study compliance to referral advice given at BC SCOREs. Gans et al.⁸ found that only 48% of PHHP SCORE participants with BC levels ≥ 240 mg/dL said that they had visited a physician four to seven months after a BC SCORE.

Methods to improve referral compliance have been investigated for other medical conditions, such as hypertension, with

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resultant 11% to fourfold increases in referral compliance rates and overall compliance rates as high as 85%.³⁴⁻⁴³ Building on the lessons learned in the previous studies, PHHP designed and conducted a study to affect compliance to referral recommendations given at BC SCOREs.

METHODS

Study population. Study subjects were recruited through SCOREs conducted at worksites ($n = 19$) and community sites ($n = 18$) throughout Rhode Island from May 1990 to June 1991. The PHHP SCORE has been described in detail elsewhere.^{11,32,33} Briefly, the SCORE included registration; total BC and weight measurement; Rate Your Plate, a self-administered eating pattern assessment;⁴⁴ a brief one-on-one counseling session, including dietary and referral recommendations according to NCEP guidelines;^{3,8} and distribution of a self-help nutrition kit. Participants took home a blood cholesterol result form with their BC level and NCEP referral recommendations clearly indicated, a cholesterol pocket record card, the Rate Your Plate questionnaire with specific behavior goals written on it, and the nutrition kit. Participants who required a referral, but had no regular physician, were given the name of a local physician to call from a physician referral list. BC levels were determined using the Reflotron® (Boehringer Mannheim Diagnostics).⁴⁵⁻⁴⁸ To ensure accuracy and precision of BC measurements, PHHP practiced stringent internal and external quality control procedures according to national guidelines.^{8,49} SCOREs were staffed by trained and certified staff and volunteers who attended an intensive training and certification program in BC measurement and summary and referral counseling.⁵⁰ Management-level PHHP staff supervised SCORE staff in the field and conducted on-site quality control checks of measurement and counseling procedures.

SCORE participants who met all of the following criteria were eligible for this study: (1) the participant's BC level was ≥ 240 mg/dL or 200–239 mg/dL with two other CHD risk factors (male gender, self-reported history of diabetes or hypertension, smoking, positive family history of premature CHD, and excess weight—greater than 20% overweight); (2) the participant had not previously been told that his or her BC level was high; (3) the participant identified a personal physician; and (4) the participant read and signed the informed consent form. The informed consent form included a request for permission to inform the participant's physician about his or her BC level and to contact the physician to request BC-related information from the participant's medical record. Of the 1965 adults screened, 45% had BC levels less than 200 mg/dL, 32% had no regular physician, 19% were already aware of a high BC level, 17% had BC levels between 200 and 239 mg/dL but less than two risk factors, and 6% refused to participate. The remaining 207 subjects were randomized to the four intervention conditions.

Interventions. Eligible study subjects were randomized into one of four groups: participant intervention (PT), physician intervention (PH), both interventions (PTPH), or usual care (UC). Within a month of the SCORE, subjects in the PT condition received a mailed personalized letter that included their BC level, a reminder to see their physician, a list of the specific lifestyle goals the subject set at the SCORE, and a PHHP magnet for them to use to hang the letter on the refrigerator as an environ-

mental prompt. During the same time frame, physicians of subjects in the PH condition received a mailed packet including a letter stating that their patient had been referred on the basis of their BC level and CHD risk factors and listing the lifestyle goals the subject set at the SCORE. The packet also included a copy of the NCEP guidelines⁵¹ and a reminder postcard (addressed to the subject), which physicians could mail to the subject. Both the participant and physician interventions were administered to group PTPH. Group UC received no further intervention and served as a comparison group.

Outcome measures. Beginning four months after the SCORE, trained interviewers administered a 12-minute structured telephone questionnaire to all study subjects, using a computer-assisted telephone interviewing (CATI) system. Recall of the physician referral was determined by response to the following question (Q8): "When you had your cholesterol measured, were you told to see your physician within the next few months?"

To determine physician referral compliance, we asked all subjects whether they had visited or phoned their physician since the BC screening (Q20). As other studies have done,^{10,11,15,16,18,20,22,25} we considered any physician visit as constituting compliance. We also asked the subjects who did contact their physicians what their physician said about their BC level. Recall of dietary recommendations was assessed by asking the following question (Q2): "When you had your cholesterol measured, were you given recommendations to change your diet or eating pattern?" To determine compliance, subjects were asked if they had made any changes in their diet or eating pattern since the SCORE (Q10). Similar questions (Q5 and Q15) were asked of all subjects to ascertain recall of and compliance to lifestyle recommendations (i.e., quit smoking, lose weight, increase physical activity).

Analysis. Analysis of variance (ANOVA) models were constructed, and chi-square tests were used to assess the equivalence of subjects in the four conditions for continuous and categorical variables, respectively. Variables considered included demographics (age, gender, education, ethnicity), self-reported risk factors (diabetes, hypertension), and measured risk factors (total BC and body mass index [BMI]). Similar analyses were performed to compare responders to nonresponders. Bivariate analyses were performed to assess the relationship between covariates and outcomes. Multiple logistic regression models were developed to assess the impact of each intervention group relative to the UC group, adjusted for extraneous factors identified in the bivariate analyses. Additional multiple logistic regression models were developed to identify predictors of recall and compliance. Relative risks (RRs) and 95% confidence intervals (CIs) were calculated. Potential confounders or predictors included gender, age, marital status, educational level, Portuguese ethnicity, possession of medical insurance that paid for physician visits, time from SCORE to interview, family history of premature heart disease, total BC level, self-reported history of diabetes and high blood pressure. All analyses were performed using the SAS system V.6.08 for Windows.⁵² The α level used for statistical tests was 0.05.

RESULTS

Trained interviewers successfully contacted 84% of the study subjects to administer the follow-up telephone interview ($n = 173$). Only five individuals refused to answer the survey. The rest

Table 1. Demographic characteristics and risk factor profile of study participants by intervention condition and overall

	Usual care (n = 45)	Patient (n = 42)	Physician (n = 39)	Patient and physician (n = 47)	Overall (n = 173)
Demographics					
Male (%)	55.6	54.8	53.3	55.9	54.9
Mean age $\pm$ standard deviation (SD)	53.9 $\pm$ 14.9	51.1 $\pm$ 13.5	50.1 $\pm$ 17.5	50.3 $\pm$ 14.6	51.4 $\pm$ 15.1
Married (%)	75.6	73.8	68.4	69.6	71.9
White (%)	95.6	95.1	94.9	100.0	96.5
Portuguese (%)	29.5	26.8	15.4	26.1	24.7
Education (%)					
< High school	33.3	17.1	25.6	29.8	26.7
HS-some college	51.1	61.0	56.4	46.8	53.5
$\geq$ College graduate	15.6	21.9	18.0	23.4	19.8
Median household income	\$35,000	\$45,000	\$45,000	\$35,000	\$35,000
Medical insurance covering MD visits (%)	60.0	62.5	60.5	55.3	59.4
Risk factor profile					
Cigarette smoking (%)	13.6	27.9	28.9	29.8	25.3
High blood pressure (%)	31.8	30.9	38.5	40.4	35.5
Diabetes (%)	2.2	2.4	5.3	8.5	4.7
Family history CHD (%)	22.2	21.4	23.7	19.6	21.6
Mean cholesterol mg/dL $\pm$ SD	245 $\pm$ 31.1	241 $\pm$ 25.5	239.8 $\pm$ 30.0	241.3 $\pm$ 24.3	242 $\pm$ 27.6
BMI $\pm$ SD	27.2 $\pm$ 4.3	27.1 $\pm$ 4.1	28.1 $\pm$ 5.1	27.2 $\pm$ 5.1	27.1 $\pm$ 4.6

had disconnected phone numbers, had moved, or were unable to be reached within a year. Neither mean time between initial SCORE and telephone interview (average = 5.5 months, range 4–12 months) nor response rates differed by treatment group.

Nonresponders to the survey were significantly younger than responders (mean age 44.4 years versus 51.4 years, $t = -2.47$, $P < .02$), and nonresponders were somewhat more likely to be smokers (41.2% versus 25.3%, $\chi^2 = 3.6$, $P = .06$). Responders and nonresponders were similar on all other variables. For the 173 respondents, there were no differences between treatment groups in baseline demographic characteristics and prevalence of risk factors by intervention group, as shown in Table 1.

Table 2 shows recall of and compliance to physician referral, dietary recommendations, and lifestyle recommendations for each intervention group. Overall, 58% of subjects recalled the physician referral recommendation. This recall differed by group condition, with greater recall occurring in group PT and group PTPH than in group PH or group UC ($\chi^2 = 15.6$, $P = .001$). Of those receiving the participant intervention (PT plus PTPH), 71% recalled the physician referral recommendation versus 45% of those who did not receive that intervention (UC plus PH). Overall, 86% of subjects recalled receiving dietary recommendations. Recall differed by group condition, with greater recall occurring in group PT and group PTPH than in group PH or group UC ($\chi^2 = 9.3$, $P = .025$). Of those receiving the participant intervention (PT plus PTPH), 93% recalled the dietary recommendations, versus 77% of those who did not receive the participant intervention (UC plus PH). The same trend occurred in the recall of lifestyle recommendations, although statistical significance was not achieved. The percentage of subjects reporting compliance to the recommendations (physician referral, dietary or lifestyle recommendations) did not differ by intervention condition. Overall, 58%, 67%, and 34% of subjects reported com-

plying to physician referral, dietary recommendations, and lifestyle recommendations, respectively. Of the subjects who reported contacting their physician after the SCORE, only 16 reported that they did *not* tell their physician about the BC screening. Study outcomes did not change when these subjects were excluded from analyses.

Table 3 shows the adjusted relative risks (RRs) relating compliance to physician referral, dietary recommendations, and lifestyle recommendations to the intervention groups. RRs shown have been adjusted for potential confounders. Subjects receiving both the PT and the PH intervention were five times more likely than the UC group to recall the physician referral. Although the trend did not achieve statistical significance, subjects in the PT group were twice as likely to recall the physician referral. Subjects receiving the PT intervention were six times more likely to recall dietary recommendations than the UC group. Neither the PT intervention nor the PH intervention increased compliance to physician referral, dietary recommendations, or lifestyle recommendations.

Table 4 presents the factors that predicted compliance to physician referral, dietary recommendations, and lifestyle recommendations. Separate multiple logistic regression models were constructed for each outcome. Predictors of compliance to physician referral included time from SCORE to interview, history of hypertension, and possession of medical insurance covering physician visits. Subjects who had more time between their SCORE and their telephone interview were more likely to visit their physician. Each additional month between SCORE and interview was associated with a 30% increase in compliance. The mean time interval for those who reported visiting their physician was approximately 5.8 months, whereas the time interval for those who did not visit their physician was approximately 5.1 months ($t = 2.99$, $P < .005$). Subjects with a history of hypertension

Table 2. Percentage of responders who recalled being given and reported actually following referral, dietary, and lifestyle recommendations by intervention condition (patient, physician, and patient and physician) and overall

	Patient (n = 42) %	Physician (n = 39) %	Patient and physician (n = 47) %	Usual care (n = 45) %	Overall (n = 172) %
Referral					
Recalled being told	61.9	38.5	78.7	51.1	58.4**
Complied	57.1	53.9	57.5	62.2	57.8
Dietary recommendations					
Recalled being told	95.2	79.5	91.5	75.6	85.5*
Complied	64.3	61.5	74.5	66.7	67.1
Lifestyle recommendations					
Recalled being told	69.1	46.2	61.7	57.8	59.0
Complied	35.7	35.9	36.2	26.7	33.5

* $P < 0.5$.

** $P < .001$.

Table 3. Impact of participant and physician interventions on participant recall of and compliance with PHHP recommendations

	Patient ^a RR (95% CI)	Physician ^a RR (95% CI)	Patient and physician RR (95% CI)
Recall			
Physician referral	1.9 (0.7, 5.6)	0.6 (0.2, 1.8)	5.0* (1.0, 25.2)
Dietary recommendations	6.2* (1.0, 37.8)	1.1 (0.3, 4.0)	1.1 (0.1, 12.1)
Lifestyle recommendations	1.6 (0.6, 4.4)	0.7 (0.3, 1.8)	1.1 (0.3, 4.4)
Compliance			
Physician referral	0.8 (0.3, 2.2)	0.7 (0.2, 1.8)	1.3 (0.3, 5.4)
Dietary recommendations	1.2 (0.4, 3.1)	1.0 (0.4, 2.6)	1.7 (0.4, 6.8)
Lifestyle recommendations	1.4 (0.5, 3.6)	1.5 (0.6, 4.2)	0.7 (0.2, 2.6)

PHHP = Pawtucket Heart Health Program.

Relative risks and 95% confidence intervals adjusted for age, sex, education, Portuguese ethnicity, marital status, smoking status, family history of premature CHD, diabetes, total blood cholesterol, elevated blood pressure, medical insurance covering MD visits, and time from SCORE to interview.

^aReference group was the usual care group.

* $P < .05$.

were 2.6 times more likely to have visited their physician after the SCORE than those without hypertension. In addition, 100% of subjects with diabetes indicated that they saw their physician after the SCORE. Those with medical insurance covering physician office visits (either fully or partially) were twice as likely to have complied than those without this medical coverage ($P = .058$).

Statistically significant predictors of dietary compliance included age and total BC level. People with baseline BC levels ≥ 240 mg/dL were three times more likely to have complied with dietary recommendations than those with BC levels between 200 and 239 mg/dL. Increasing age was inversely related to dietary compliance, with each one-year increase in age corresponding to a 3% decrease in compliance. Although not quite achieving statistical significance, the following factors did show a trend in predicting dietary compliance. History of hypertension and medical insurance for physician visits were positively related to dietary compliance, and cigarette smoking was inversely related to compliance with dietary recommendations.

There were no statistically significant predictors of lifestyle compliance; however, there was a trend with increasing age inversely related to compliance and baseline BC ≥ 240 mg/dL positively related to compliance.

DISCUSSION

Referral compliance. Overall, 58% of study subjects recalled that they were told at the SCORE to contact their physician. This recall rate was significantly higher than rates of 22%–49% found in previous studies.^{15,24} However, because of NCEP guidelines and current study criteria, 100% of the study subjects should have received physician referral recommendations. Because of the rigorous PHHP training, certification, and quality assurance program, and the fact that study staff had several years of experience conducting BC screenings, we believe that SCORE staff members did give the referral recommendations. In addition, all SCORE participants received a take-home chole-

terol result form with the referral recommendations clearly indicated. Therefore, failure to recall the referral is most likely due to subjects' faulty memories. The PT intervention apparently improved recall of physician referral recommendations, but the PH intervention did not, most likely because physicians did not mail the reminder postcards. In fact, only 20% of subjects in this group said they received the physician postcard, whereas 69% of PT subjects said they received the participant letter. Why so few physicians apparently mailed the postcard remains unclear. Perhaps physicians do not trust public BC screenings because of bad press in the early 1990s about public screenings.⁵³ Alternatively, physicians may have decided to discuss the patient's cholesterol at a future scheduled appointment rather than schedule a special appointment for cholesterol only.

Neither the PT intervention nor the PH intervention had any statistically significant impact on referral compliance. Overall, 58% of subjects reported actually contacting their physician after the SCORE. This rate of referral compliance is higher than that found in an earlier PHHP study,¹⁵ which found a referral compliance rate of 48% for individuals with BC levels ≥ 240 mg/dL. However, the previous study included subjects who did not have physicians, which should have lowered referral compliance rates. Other studies have shown referral compliance rates ranging from 17% to 63%, similar to or lower than the rates found in this study.^{16,17,19,20,21,23,28-31} However, several previous studies have shown higher referral compliance rates after BC screening ranging from 65% to 79%.^{13,14,22,24-27} These latter studies cannot be appropriately compared to this study because they included individuals with higher BC levels than those in this study, had low response rates to the follow-up survey reflecting nonresponse bias, asked the referral compliance question only to those who recalled referral recommendations, or had a longer follow-up time (greater than one year). In this study, the most common reasons given by the 42% of subjects who did not contact their physician were no need/BC OK (25%), have a future appointment scheduled (21%), no time (12%), feel OK/healthy (12%), doing things on my own to lower BC (11%), and no one told me to see my physician (7%).

In this study, the longer the time interval between SCORE and survey, the more likely the subject had contacted his or her physi-

cian. Perhaps subjects needed a longer time to schedule an appointment and actually see a physician. Many of the other studies that found even lower referral compliance rates (17%–46%) had a shorter follow-up time (one to three months).^{18,20,28,31} In addition, Havas et al.²⁸ found a trend with increasing referral compliance as time interval after screening increased (two to four months = 51.5%, five to six months = 58.1%, six to 12 months = 65.6%). Perhaps the two-to-three-month time frame recommended by the NCEP^{3,8} for individuals to see their physician after a BC screening is not realistic, given the amount of time needed for individuals to schedule and keep "nonacute" physician visits. Alternatively, individuals may not be willing to spend the time and money to schedule a special appointment about their BC when they have a future physician appointment scheduled. In fact, in this study 21% of subjects who had not contacted their physician by the time of the telephone survey indicated having a future appointment scheduled.

Having medical insurance that paid for physician visits was positively associated with referral compliance. If subjects do not have medical insurance for physician visits, logically, they will be less likely to visit their physician for "nonacute" reasons. In addition, they may be more likely to attend BC screenings and health fairs, where they can get health information less expensively than in the physician's office, and to try to "do things on their own" to lower BC rather than seek treatment from a physician. Similarly, in a study of cancer referrals, Lane found that "not having to pay for medical services" predicted referral compliance.⁵⁴

In this study, presence of other health problems (i.e., hypertension or diabetes) also predicted whether a subject contacted a physician after the BC SCORE. Since individuals with these health problems are probably already seeing a physician regularly, they could easily add a discussion about cholesterol to another visit without incurring additional time and cost. In a study of referrals for hypertension, McClellan et al. also found that presence of comorbid conditions predicted referral compliance.⁵⁵

Dietary compliance. Overall, 86% of subjects recalled that they were told at the SCORE to make dietary changes. This recall rate is higher than the 75% recall rate found in a previous

Table 4. Factors predicting compliance to physician referral, dietary, and lifestyle recommendations

	Physician referral RR (95% CI)	Dietary recommendations RR (95% CI)	Lifestyle recommendations RR (95% CI)
Time since SCORE (each additional month)	1.3** (1.0, 1.7)	— ^a	
Age (each additional year)	— ^a	0.97** (0.9, 1.0)	0.98* (0.95, 1.00)
Cholesterol ≥ 240 mg/dL	— ^a	3.3*** (1.4, 7.3)	1.8 (0.9, 3.8)
Cigarette smoker	— ^a	0.43* (0.2, 1.03)	— ^a
History of hypertension	2.6** (1.1, 5.8)	2.10* (0.9, 4.9)	— ^a
Medical insurance for MD visits	2.1* (0.98, 4.4)	1.89 (0.8, 4.2)	— ^a

Relative risks and 95% confidence intervals adjusted for age, sex, education, Portuguese ethnicity, marital status, smoking status, family history of premature CHD, total blood cholesterol, elevated blood pressure, medical insurance covering MD visits, and time from SCORE to interview.

^aOnly terms whose *P* values were less than .2 were included in this table.

**P* < .10.

***P* < .05.

****P* < .01.

PHHP study.¹⁵ Recall of dietary advice did differ by intervention group, with those who received the PT and PTPH interventions significantly more likely to recall being told to change their diet. Again, the PH intervention appeared to have had no influence on recall rates, likely because subjects did not receive the postcard. More study subjects recalled dietary advice than physician referral advice (85% versus 58%), perhaps because the majority of time during the counseling session was spent on the dietary education portion. In addition, the dietary discussion occurred towards the end of the session, whereas the discussion of referral recommendations occurred at the beginning of the session. Further, subjects were actively involved in setting dietary goals but were just told the referral recommendation.

Overall, 67% of subjects self-reported making dietary changes. This rate is similar to that found in a previous PHHP study.¹⁵ Neither the PT intervention nor the PH intervention improved dietary compliance. Subjects with higher BC levels were significantly more likely to self-report making dietary changes than those with lower BC levels. This result may have occurred because subjects with higher BC levels had "less healthful" diets at baseline, and thus had more room to make dietary changes. Or perhaps they were more motivated to make dietary changes because of their high BC level or were more likely to exaggerate the changes they made because they knew they should have made dietary changes. Other studies have found 40% to 90% of BC screening subjects reporting dietary changes at follow-up,^{14,15,18,20,22,24,26,27,29,31,56} with higher rates of dietary change associated with higher BC levels and low response rates perhaps reflecting the responses of a self-selected, more motivated group. We also found that older individuals were less likely to report making dietary changes. These individuals may have had more healthful diets at baseline, or perhaps it was more difficult for them to make changes because of a set eating pattern, expense, etc. Educational approaches at BC screenings may need to focus especially on participants with borderline-high BC levels and older adults.

Lifestyle compliance. Overall, 94% of subjects had at least one risk factor that required lifestyle change (smoking, excessive weight, or lack of exercise), yet only 59% of subjects recalled being given lifestyle change recommendations at the SCORE. This percentage is similar to the 54% found previously.¹⁵ The PH and PT interventions had no impact on recall of or compliance to lifestyle recommendations, although a trend suggested that the groups receiving the participant letter (PT and PTPH) may have been more likely to recall being told to make lifestyle changes.

Approximately one third of subjects reported actually making other lifestyle changes after the SCORE. This percentage is similar to the 37% found previously by PHHP.¹⁵ Even though lifestyle change was not the major focus of the counseling session, 14% of smokers said they quit, 9% of overweight individuals said they lost weight, and almost one quarter of nonexercisers reported beginning to exercise after the SCORE. Other studies have also found lifestyle changes including weight loss, smoking cessation, and an increase in exercise after BC screening programs.^{14,15,22,24,26,31}

Summary and Implications

One limitation of this study was the relatively small sample size in each intervention condition. However, the absence of any consistent trends suggesting enhanced compliance most likely rules

out any large intervention effects. Therefore, we conclude that the PT and PH interventions used in this study are not likely to enhance significantly compliance to referral or dietary or lifestyle recommendations. The participant intervention successfully improved subjects' recall of referral and dietary recommendations, but that was not enough to affect behavior. Interventions similar to those used in this study have successfully enhanced referral compliance of individuals attending hypertension screenings,³⁶ but were not successful with BC screening subjects in this study. However, even though the previous study improved the referral compliance rate with intervention, the overall compliance rate was only 49%.³⁶ In this study, the overall referral compliance rate was 58%, plus another 12% with future appointments scheduled. Possibly, the SCORE itself, which includes counseling and take-home materials as well as screening, already approached the maximum referral compliance rate possible in a screening setting, so the PH and PT interventions were unable to further increase compliance. Others have suggested that the educational component of a BC screening program may actually counteract physician referral advice by arming subjects with strategies for attempting to reduce BC levels on their own.³¹ This possibility needs to be studied further.

To improve referral compliance rates at cholesterol screenings, perhaps more time should be spent discussing the referral or the referral discussion should become more interactive, with subjects signing a contract to see their physician or verbally repeating referral advice at the end of the SCORE. Research on methods to improve compliance with appointment-keeping for medical services has found these interventions successful.⁵⁷ In addition, more frequent contact may be needed after the screening, or perhaps a different type of intervention may be necessary to improve referral compliance. Jones et al. have had much success using a Health Belief Model intervention^{34,41,42} to encourage emergency room patients with a variety of medical problems to comply with referral advice. Methods such as multiple mailings, incentives, video intervention, or telephone reminders may also be helpful.^{37,38} Future research should also study the referral compliance rates of individuals who lack a regular physician and whose compliance rates are probably even lower than those in this study. Using a physician referral list and focused interventions may be necessary to encourage referral compliance for these individuals.

BC screening and education programs can be valuable tools for identifying individuals with elevated BC levels, educating them about how to lower their BC level^{19,25,26,56,59-62} and encouraging lifestyle changes to reduce other CHD risk factors.^{10,11,18,20,30,27} However, BC screening and education programs may need to imitate their forefathers, the hypertension screening programs,⁶³⁻⁶⁵ and begin to focus less on screening and more on control, referral, and follow-up.

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